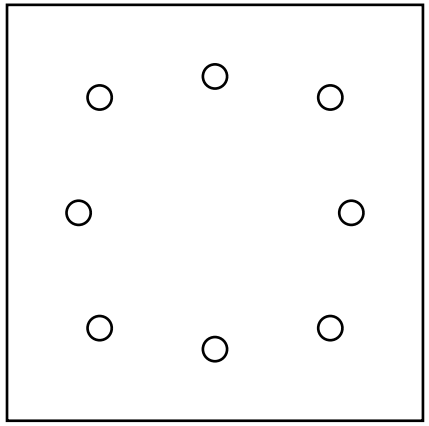
M01 / AIO adapter

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION



FR4 t=1.6mm; quantity 1

Width 55 x height 55mm

Origin: centre; x right, y up.

Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

Holes: x, y, diameter THRU

1: -15.2500, -15.2500, 3.2

2: -15.2500, 15.2500, 3.2

3: 15.2500, -15.2500, 3.2

4: 15.2500, 15.2500, 3.2

5: 18.0312, 0.0000, 3.2

6: -18.0312, 0.0000, 3.2

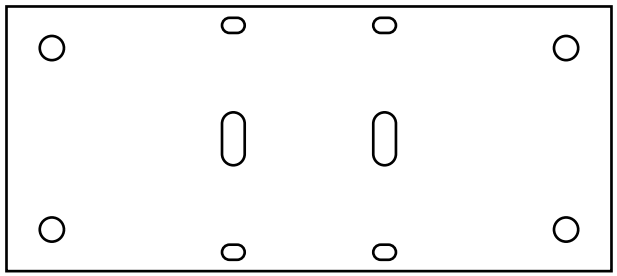
7: 0.0000, 18.0312, 3.2

8: 0.0000, -18.0312, 3.2

Frame 30.5 square; AIO 25.5 square rotated 45deg. Arrow FORWARD, not board edge. HOLD H01.

M02 / Bus carrier

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION



FR4 t=1.6mm; quantity 1

Width 80 x height 35mm

Origin: centre; x right, y up.

Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

Holes: x, y, diameter THRU

1: -34.0000, -12.0000, 3.2

2: -34.0000, 12.0000, 3.2

3: 34.0000, -12.0000, 3.2

4: 34.0000, 12.0000, 3.2

S1: (-10,0) overall 3 x 7

S2: (10,0) overall 3 x 7

S3: (-10,-15) overall 3 x 2

S4: (-10,15) overall 3 x 2

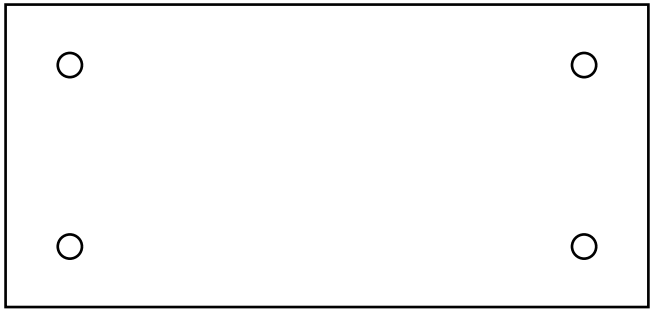
S5: (10,-15) overall 3 x 2

S6: (10,15) overall 3 x 2

Rail centres y=+/-8.7625; gap8.0. Strap lanes x=+/-10 clear solder zones. Edge strength HOLD H02.

M03 / Bus top guard

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION



PC t=1.0mm; quantity 1

Width 85 x height 40mm

Origin: centre; x right, y up.

Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

Holes: x, y, diameter THRU

1: -34.0000, -12.0000, 3.2

2: -34.0000, 12.0000, 3.2

3: 34.0000, -12.0000, 3.2

4: 34.0000, 12.0000, 3.2

12mm nylon spacers; side skirts and protected wire exits still required by H02. This top sheet is NOT a complete guard.

M04 / C110 bus rail

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION

C110 t=3.175mm; quantity 2

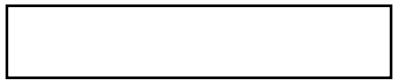
Width 50.8 x height 9.525mm

Origin: centre; x right, y up.

Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

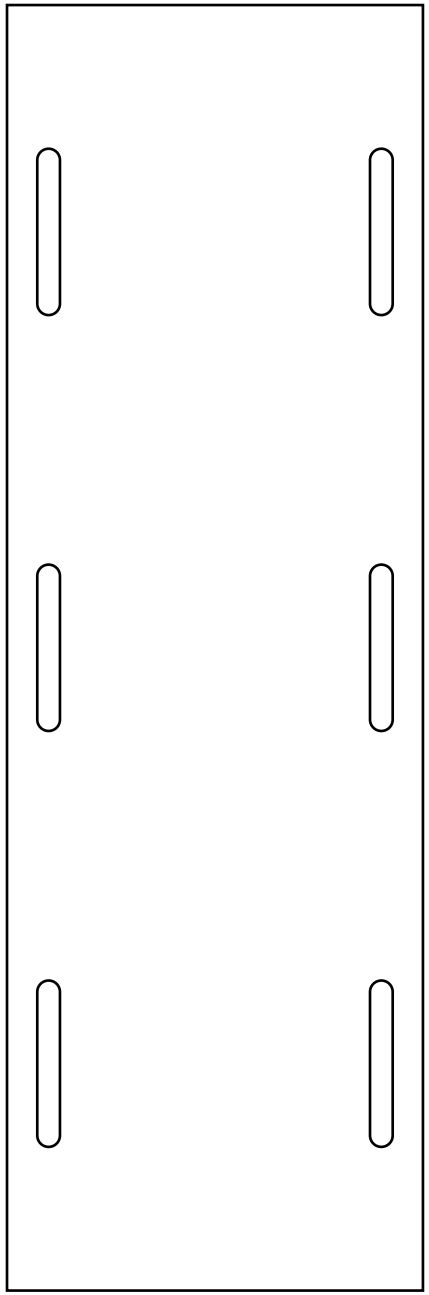
Holes: x, y, diameter THRU



No drilled current joints; deburr R0.3; solder zones x=+/-16..25.4. Polarity permanently marked. HOLD H02.

M05 / Battery cradle blank

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION



FR4 t=1.6mm; quantity 1

Width 55 x height 170mm

Origin: centre; x right, y up.

Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

Holes: x, y, diameter THRU

S1: (-22,-55) overall 3 x 22

S2: (-22,0) overall 3 x 22

S3: (-22,55) overall 3 x 22

S4: (22,-55) overall 3 x 22

S5: (22,0) overall 3 x 22

S6: (22,55) overall 3 x 22

Three20mm straps through paired22mm slots. Slider/frame mount holes OMITTED: H03, not a completed cradle.

M06 / Battery end stop

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION

FR4 t=1.6mm; quantity 2

Width 55 x height 15mm

Origin: centre; x right, y up.

Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

Holes: x, y, diameter THRU

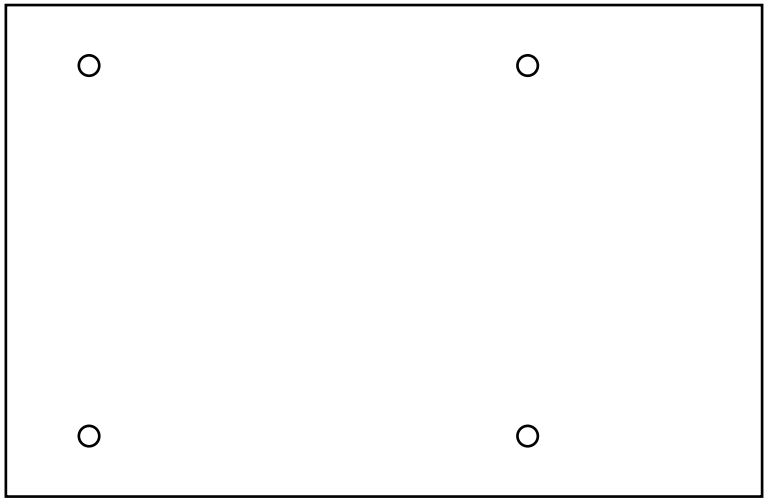
1: -18.0000, 0.0000, 3.2

2: 18.0000, 0.0000, 3.2

Vertical stop; bolt to cradle via insulating angles, not unsupported adhesive. HOLD H03.

M07 / Pi insulating base

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION



PC t=0.6mm; quantity 1

Width 100 x height 65mm

Origin: centre; x right, y up.

Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

Holes: x, y, diameter THRU

1: -39.0000, -24.5000, 2.7

2: -39.0000, 24.5000, 2.7

3: 19.0000, -24.5000, 2.7

4: 19.0000, 24.5000, 2.7

Pi PCB centred;58x49 holes offset x=-10 from PCB centre. Exact connector/part overlay HOLD H04.

M08 / Pi longitudinal rail

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION

PC t=1.2mm; quantity 2

Width 80 x height 8mm

Origin: centre; x right, y up.

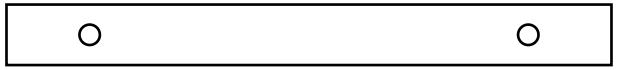
Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

Holes: x, y, diameter THRU

1: -29.0000, 0.0000, 2.7

2: 29.0000, 0.0000, 2.7



Rail centres on base x=-10,y=+/-24.5. Clamp through Pi posts. HOLD H04.

M09 / Fan bridge

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION

PC t=0.8mm; quantity 1

Width 40 x height 30mm

Origin: centre; x right, y up.

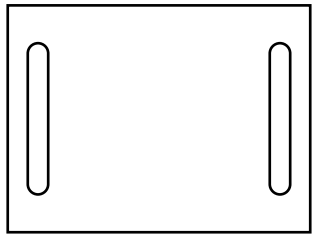
Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

Holes: x, y, diameter THRU

S1: (-16,0) overall 2.7 x 20

S2: (16,0) overall 2.7 x 20



Fan opening and mounting pattern TRANSFER after exact3368 drawing/part. Do not obstruct blades. HOLD H04.

M10 / Camera adapter blank

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION

PC t=0.8mm; quantity 1

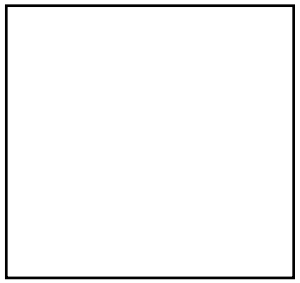
Width 38 x height 36mm

Origin: centre; x right, y up.

Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

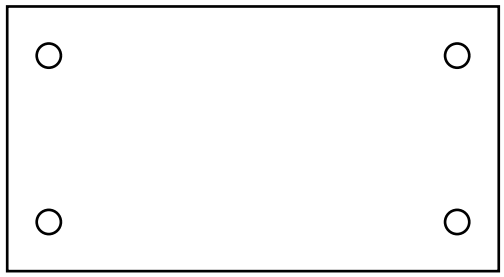
Holes: x, y, diameter THRU



Camera3 and pan/tilt fastening patterns OMITTED pending authoritative overlay; not a finished adapter. HOLD H05.

M11 / Front accessory shelf

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION



FR4 t=1.0mm; quantity 1

Width 65 x height 35mm

Origin: centre; x right, y up.

Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

Holes: x, y, diameter THRU

1: -27.0000, -11.0000, 3.2

2: -27.0000, 11.0000, 3.2

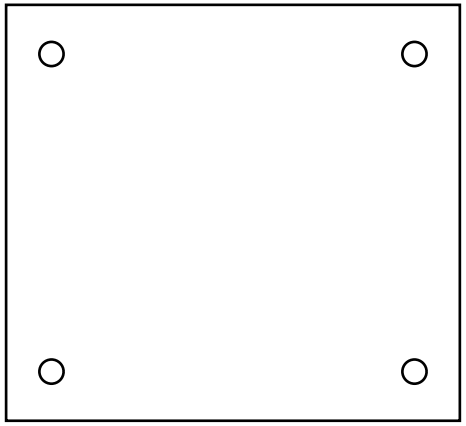
3: 27.0000, -11.0000, 3.2

4: 27.0000, 11.0000, 3.2

IR/light and pan/tilt mount holes TRANSFER; separate optics from hot IR boards. HOLD H05.

M12 / D500 base

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION



FR4 t=1.0mm; quantity 1

Width 60 x height 55mm

Origin: centre; x right, y up.

Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

Holes: x, y, diameter THRU

1: -24.0000, -21.0000, 3.2

2: -24.0000, 21.0000, 3.2

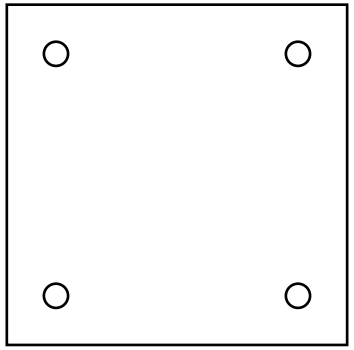
3: 24.0000, -21.0000, 3.2

4: 24.0000, 21.0000, 3.2

Mast support holes only; sensor mounting holes omitted pending dimensioned vendor drawing. HOLD H06.

M13 / GNSS platform

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION



FR4 t=1.0mm; quantity 1

Width 45 x height 45mm

Origin: centre; x right, y up.

Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

Holes: x, y, diameter THRU

1: -16.0000, -16.0000, 3.2

2: -16.0000, 16.0000, 3.2

3: 16.0000, -16.0000, 3.2

4: 16.0000, 16.0000, 3.2

Nonmagnetic restraint; exact12044 base attachment TRANSFER. HOLD H06.

M14 / TFmini base

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION

FR4 t=1.0mm; quantity 1

Width 48 x height 25mm

Origin: centre; x right, y up.

Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

Holes: x, y, diameter THRU

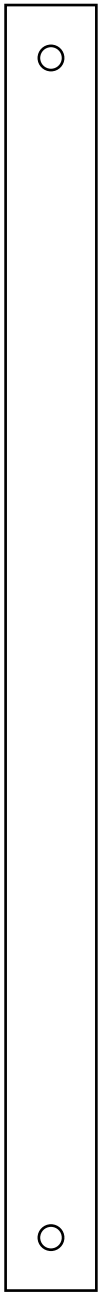
1: -19.0000, 0.0000, 3.2

2: 19.0000, 0.0000, 3.2

Sensor mounting holes omitted pending vendor overlay. Downward optical aperture unobstructed. HOLD H07.

M15 / Landing strut study

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION



FR4 t=1.6mm; quantity 4

Width 12 x height 170mm

Origin: centre; x right, y up.

Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

Holes: x, y, diameter THRU

1: 0.0000, -78.0000, 3.2

2: 0.0000, 78.0000, 3.2

Requires bracing and frame/foot joints; flat strips alone NOT released as landing gear. HOLD H08.

M16 / Landing foot

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION

FR4 t=1.6mm; quantity 4

Width 40 x height 15mm

Origin: centre; x right, y up.

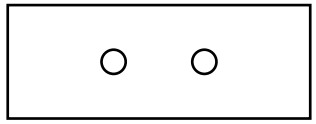
Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

Holes: x, y, diameter THRU

1: -6.0000, 0.0000, 3.2

2: 6.0000, 0.0000, 3.2



Foot pads/strut angles/bracing require resolved joint drawing H08.

M17 / Cargo lid/base

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION

PC t=1.0mm; quantity 2

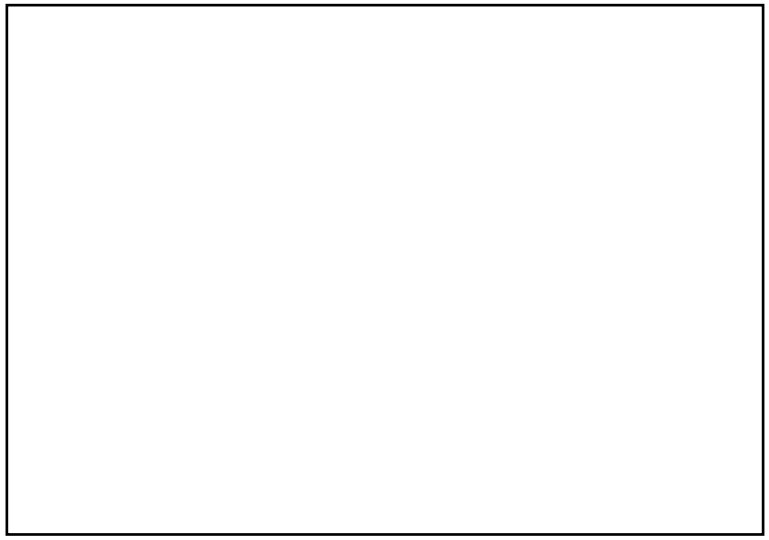
Width 100 x height 70mm

Origin: centre; x right, y up.

Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

Holes: x, y, diameter THRU



Outside planning envelope100x70x60. Joints/strike/hinge NOT released. HOLD H09.

M18 / Cargo long wall

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION

PC t=1.0mm; quantity 2

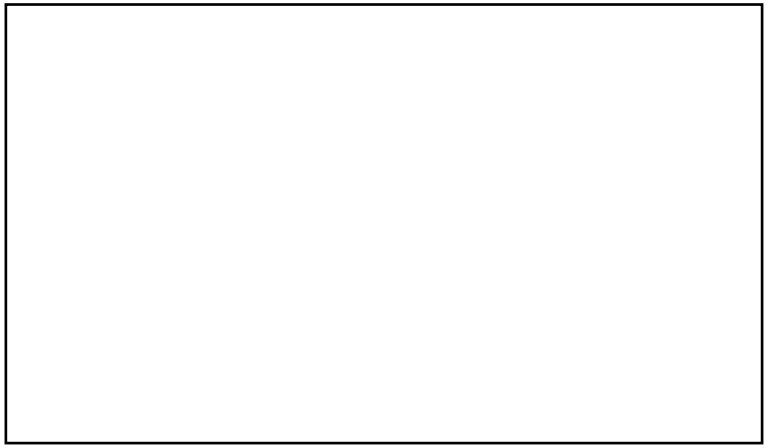
Width 100 x height 58mm

Origin: centre; x right, y up.

Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

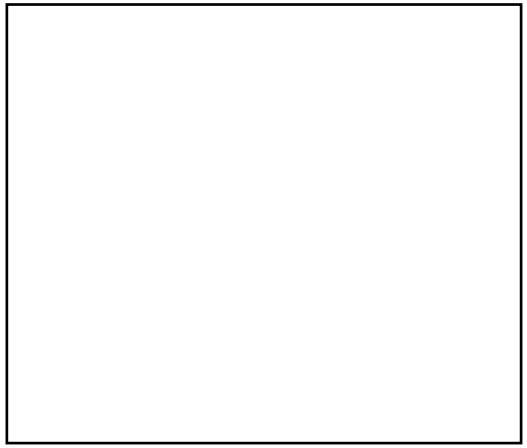
Holes: x, y, diameter THRU



Wall between lid and base; captive internal fasteners; no exterior hand-release. HOLD H09.

M19 / Cargo end wall

ENGINEERING STUDY - NOT RELEASED FOR FABRICATION



PC t=1.0mm; quantity 2

Width 68 x height 58mm

Origin: centre; x right, y up.

Outer +/-0.25; holes +/-0.10mm.

Slot radius=min(width,height)/2.

Holes: x, y, diameter THRU

Ends between long walls; corner joints and latch detail NOT released. HOLD H09.